Egg Segmentation: Classical Image Processing vs SAM Ground Truth

Repository: https://github.com/atomiclifestyle/egg_segmentation.git
Video tracking output (Google Drive): https://drive.google.com/file/d/16-N6Ag4vIMdx15ZJXwc_v4KcCdPXqyXk/view?usp=sharing

1. Summary

Task	What was done	Outcome
1	Grounding DINO + SAM pipeline to produce egg masks for every image in train/valid/test	Used as ground truth for Tasks 2-3
2	Leakage-aware 4:1 dev/eval split; classical HSV pipeline built through 15 logged trials	Dev mIoU 0.434, instance F1 0.405
3	Evaluation of the classical pipeline against the SAM masks	Eval mIoU 0.327, instance F1 0.344; most failures are wrong egg counts
4	Per-frame segmentation with a centroid-based Hungarian tracker	Partial: tracking quality is not satisfactory

2. Task 1: Ground truth with SAM

SAM cannot find objects by name, so detection and segmentation are two stages:

1. **Detect.** Grounding DINO (`grounding-dino-base`) with the text prompt "an egg." returns boxes (box threshold 0.30, text threshold 0.25). Boxes covering more than 90% of the image are dropped and the rest are de-duplicated with NMS (IoU 0.5).
2. **Segment.** Each box prompts SAM (`sam-vit-base`), giving one mask per egg.

3. Task 2: Classical segmentation

3.1 Dev/eval split

The 110 images of the train, valid and test folders were merged and split 4:1 into **88 dev** and **22 eval** images (seed 42). Splitting by image would leak information, because augmented copies of one photo (flips, rotations, brightness changes) would land on both sides:

- images sharing the same base filename (augmentation suffix removed) are in one group;
- images whose perceptual hash (dHash) differs by at most 10 bits, under any of the 8 flips/rotations, are in one group.

All tuning used the dev set only. The eval set was scored once, at the end.

3.2 Final algorithm

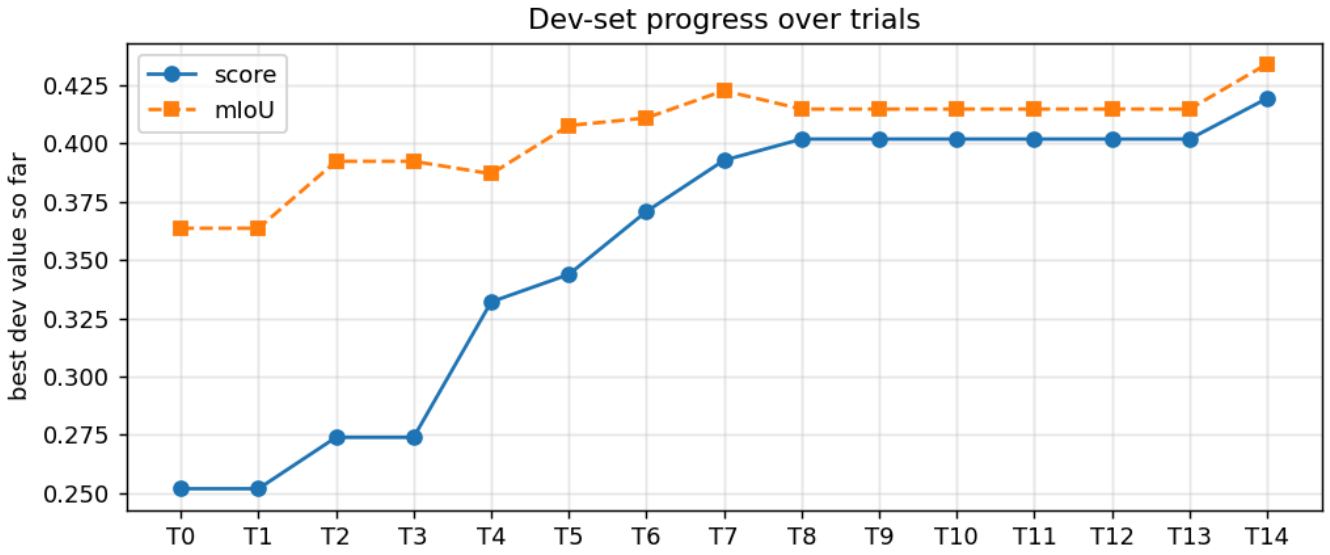
{"method": "hsv", "blur": 0.01, "s_scale": 0.6, "v_scale": 1.15, "open": 0.03, "close": 0.06, "fill": true, "min_area": 0.01, "max_area": 0.9, "solidity": 0.0, "convex": false, "watershed": false, "grabcut": true}

3.3 Experiment log (dev set only)

Each trial starts from the best configuration so far. A change is kept only if it raises the dev **score = 0.5 x (mIoU + instance F1)** by at least 0.002.

Trial	Change (parameters)	Why it was tried	mIoU	Inst. F1	Score	Decision and reason
T0	Grayscale + Otsu threshold	Simplest baseline: eggs are usually brighter than their surroundings	0.364	0.140	0.252	**Keep.** Baseline
T1	Canny edges, close gaps, fill closed contours	Egg outline is a clean closed ellipse	0.111	0.086	0.099	Discard. Far worse (-0.153)
T2	HSV: low saturation and high value	Eggs are bright and colourless; backgrounds usually are not	0.392	0.155	0.274	**Keep.** +0.022
T3	Lab: low chroma and high lightness	More perceptually uniform than HSV	0.373	0.077	0.225	Discard. -0.049
T4	Open 0.01, close 0.02, fill holes	Remove speckle; fill holes from highlights and shell spots	0.387	0.277	0.332	**Keep.** +0.058
T5	Open 0.03, close 0.06	Test whether stronger smoothing keeps helping	0.408	0.280	0.344	**Keep.** +0.012
T6	Minimum component area 0.01	Remove small false blobs	0.411	0.331	0.371	**Keep.** +0.027
T7	Saturation scale sweep (table below)	Otsu is only a heuristic split	0.423	0.363	0.393	**Keep** `s_scale` = 0.6. +0.022
T8	Value scale sweep (table below)	Shadowed side of the egg is darker	0.415	0.389	0.402	**Keep** `v_scale` = 1.15. +0.009
T9	Convex hull per egg	Eggs are convex; repair dents from shadows	0.407	0.389	0.398	Discard. -0.004
T10	Solidity filter >= 0.85	Non-egg blobs are usually non-convex	0.294	0.449	0.372	Discard. -0.030 (see below)
T11	Blur 0.03	Suppress texture before thresholding	0.403	0.371	0.387	Discard. -0.015

Trial	Change (parameters)	Why it was tried	mIoU	Inst. F1	Score	Decision and reason
T12	Watershed split, seed fraction 0.5	Separate touching eggs	0.397	0.336	0.366	Discard. -0.036
T13	Watershed split, seed fraction 0.7	Fewer over-splits than 0.5	0.370	0.352	0.361	Discard. -0.041
T14	GrabCut boundary refinement	Colour-model refinement of the egg edge	0.434	0.405	0.419	**Keep.** +0.017



{: .fig-small }

Figure 1. Best dev score so far after each trial.

4. Task 3: Evaluation and analysis

The SAM masks from Task 1 are the ground truth and the output of the final pipeline is the prediction.

4.1 Quantitative results

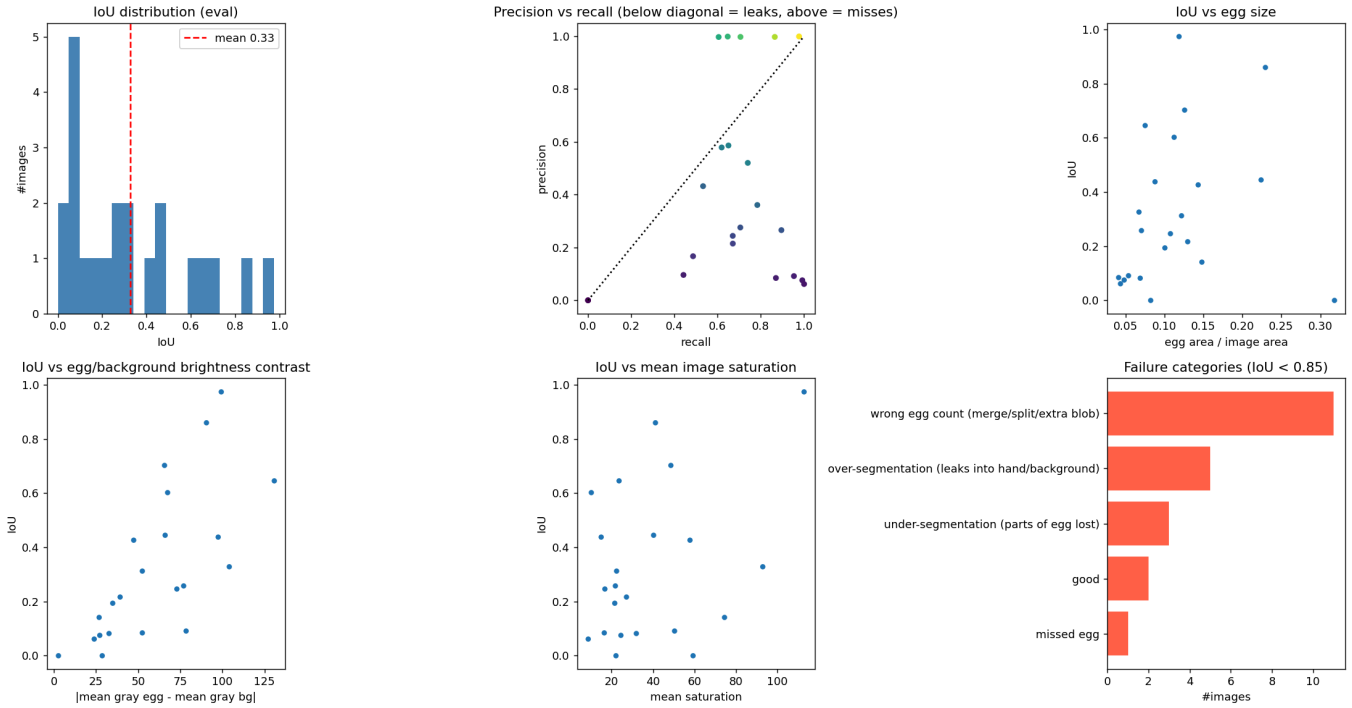


Figure 2. Eval-set analysis: IoU distribution, precision vs recall, and IoU against egg size, contrast, saturation and failure type.

4.2 Qualitative results

Best-scoring eval samples. Columns: image, SAM ground truth, prediction, error map (green correct, red false positive, blue missed).

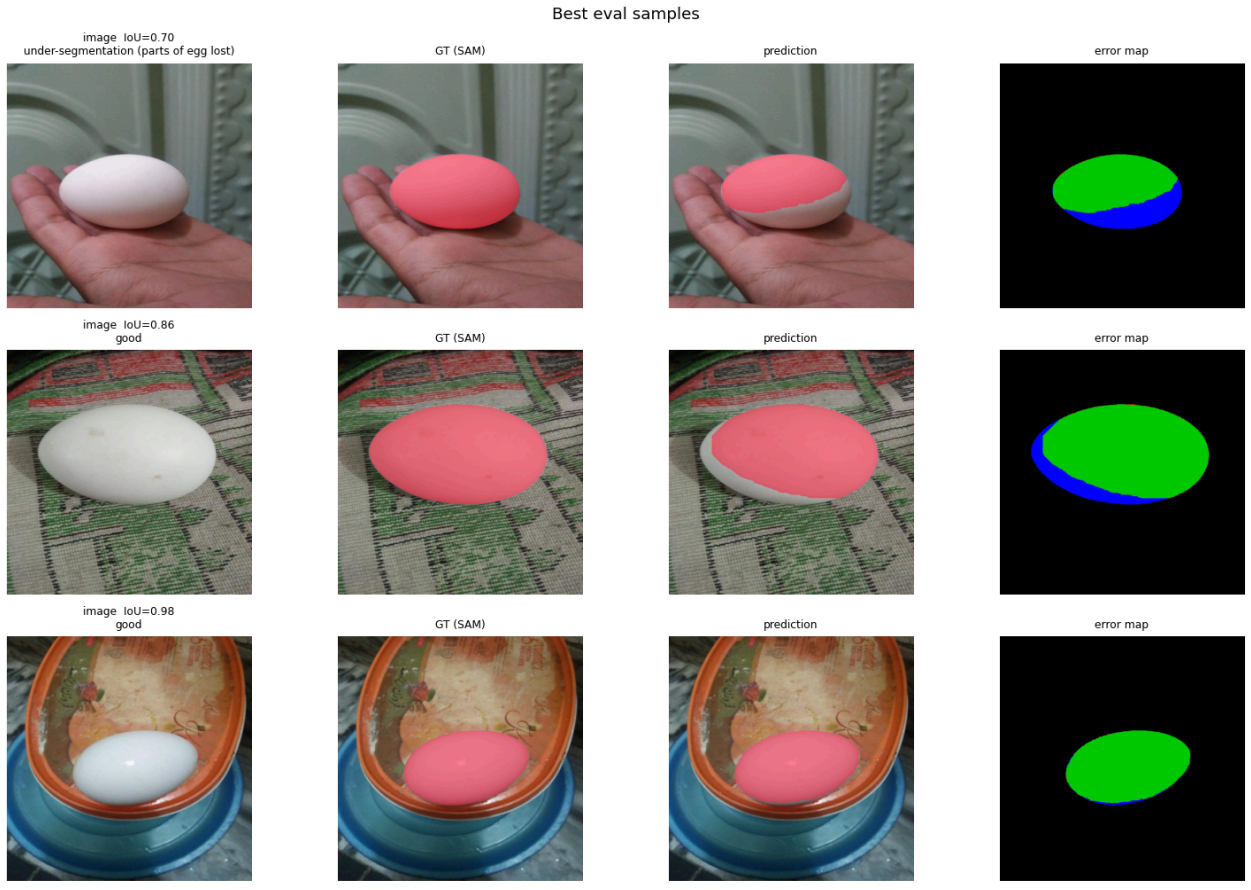


Figure 3. Best-scoring eval samples.

Worst-scoring eval samples, same layout.

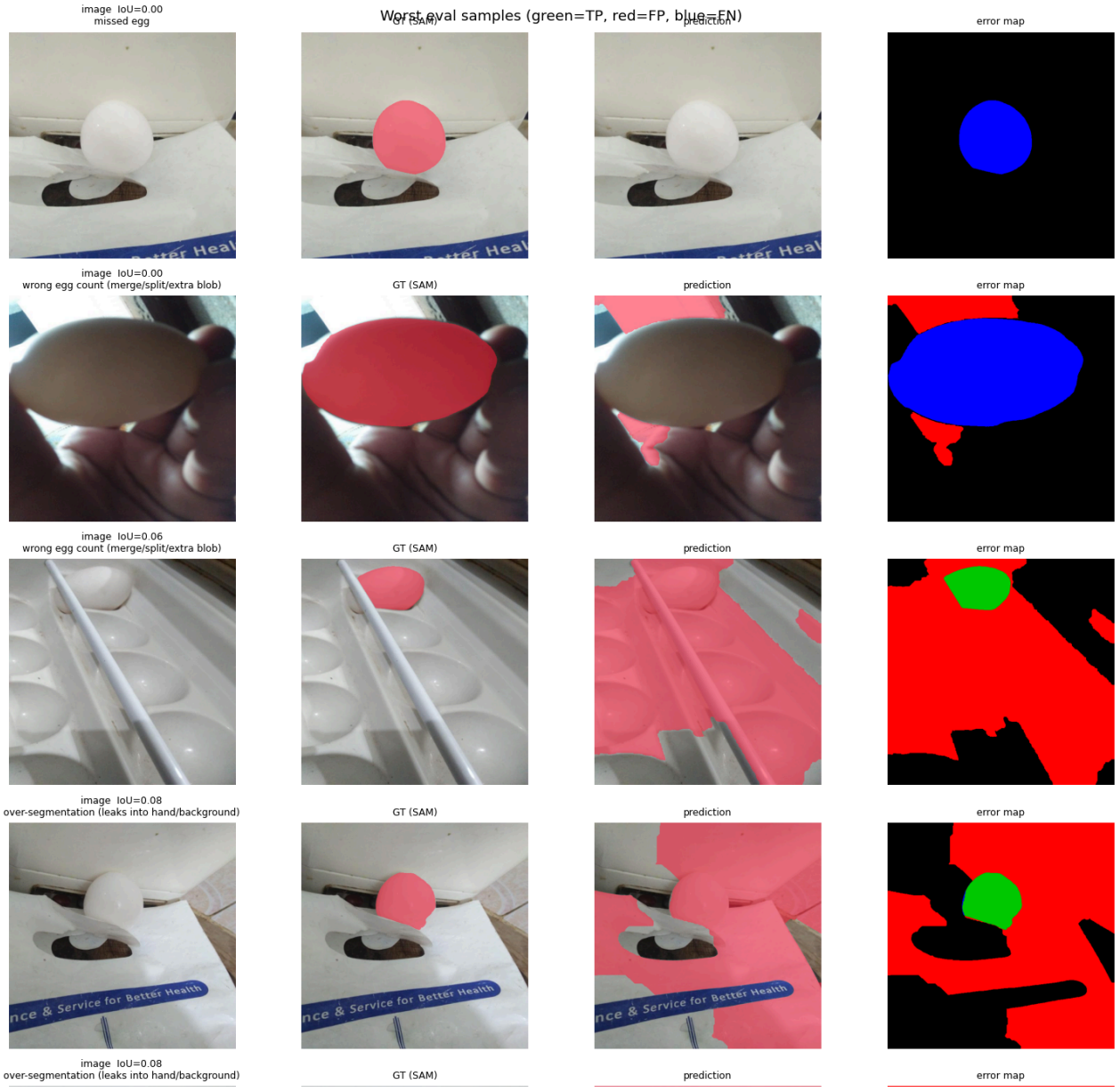


Figure 4. Worst-scoring eval samples.

4.3 Gaps and improvements

- **Per-component egg scoring.** Because wrong counts are the main failure, score each candidate region by how well it fits an ellipse (RANSAC or Hough on edges) and reject poor fits. This also targets the over-segmentation seen in the error maps.
- **Per-image colour clustering.** Cluster pixels in Lab (for example k-means with 2-3 clusters) and pick the most egg-like, convex cluster, which should be less sensitive to low egg/background contrast than a global threshold.
- **Illumination normalisation.** CLAHE or shadow removal before thresholding, to recover shadowed egg pixels and reduce the brightness dependence.
- **Wider parameter search.** Extend the `s_scale` and `v_scale` sweeps beyond their current edge values, and tune with cross-validation over dev groups rather than a single split.

- **Better splitting of touching eggs.** Watershed failed here; distance-transform peaks combined with ellipse fitting is a more constrained alternative.

5. Task 4: Video tracking (partial submission)

Method. The final segmentation pipeline runs on every frame. Each egg's centroid is computed and detections are matched to existing tracks with the Hungarian algorithm on predicted positions, using a distance gate of 12% of the frame diagonal. Each track predicts its next position with a smoothed constant velocity and survives up to 15 missed frames. A track is displayed only after 3 matched frames, to suppress one-frame false detections. The output video shows each egg's mask, a persistent ID label and a recent trail.

Status. Tracking quality is not satisfactory, so this task is submitted as partial.

Next steps. Use a stronger per-frame detector or SAM-based segmentation for video; add appearance features (colour or shape) to the matching cost; replace the constant-velocity step with a Kalman filter.

Output video: https://drive.google.com/file/d/16-N6Ag4vIMdx15ZJXwc_v4KcCdPXqyXk/view?usp=sharing